Behavioral Control Through System Prompts: A Comparative Structural Analysis of Public Governance Artifacts Across Commercial LLM Deployments

Syed Abdul Aman
Independent Researcher
amanbaba9404522@gmail.com

July 2026

Abstract

Large language model providers increasingly publish documents intended to govern model behavior at scale, ranging from training-time value specifications to runtime authority hierarchies to per-product deployed instructions. Despite substantial public interest in these artifacts — evidenced by prompt-collection repositories with tens of thousands of stars and an active empirical literature on prompt-extraction attacks [Yang et al., 2026] — direct comparative content analysis of these documents' primary text remains rare. Prior literature surveys task-oriented prompting techniques for benchmarking (which concerns user-authored prompts, not provider-authored governance), studies the security of prompt confidentiality (which concerns extraction, not content), or examines system-level instructions as an object of research and policy discourse [Neumann et al., 2026] without directly coding the primary governance documents themselves. This paper introduces a three-part taxonomy of behavioral-control artifacts — training-time constitutions, runtime authority specifications, and deployed inference-time system prompts — and applies a credibility-tiered comparative methodology across four major labs (Anthropic, OpenAI, Google DeepMind, xAI) and a 25-plus-product corpus of coding-agent tools. We find that labs diverge not only in the content of behavioral constraints but in which artifact type they use to express them at all; that transparency is sometimes reactive (incident-driven) rather than proactive; that product category (consumer chat versus coding agent) systematically predicts which instruction categories dominate a deployed prompt; and, using a publicly verifiable 18-revision changelog spanning 23 months, that at least one lab's self-description of its governance artifact as a "living document" can be independently confirmed against dated evidence rather than taken on trust. We argue this taxonomy resolves a category error common in informal prompt-comparison work, and we outline a research agenda for treating behavioral-control artifacts as first-class objects of study in AI governance and NLP.

Keywords: system prompts, LLM alignment, AI governance, prompt engineering, behavioral control, transparency

1 Introduction

Every deployed conversational AI system operates under some form of provider-authored instruction layer that sits above the user's own input — commonly called a "system prompt." As these systems have moved from research demonstrations to consumer and enterprise products, the documents that constrain their behavior have grown from short, informal instructions into substantial artifacts: Anthropic's constitution for Claude runs to roughly 23,000 words [Anthropic, 2026a]; OpenAI's Model Spec runs to 128 pages in the revision analyzed here [OpenAI, 2025].[1] At the same time, a large ecosystem of public repositories exists purely to collect and share these documents, whether officially released or reconstructed through extraction, with the single largest such repository exceeding 140,000 stars on GitHub at the time of writing [Valbuena, 2026].

Despite this level of public attention, the documents themselves have rarely been treated as a coherent object of direct comparative content analysis. Three adjacent literatures exist. The first is the prompt-engineering literature, which develops taxonomies of *task* prompts — the instructions a user or application developer supplies to elicit a specific output — for purposes of benchmarking and technique comparison [Karmaker Santu and Feng, 2023, Sahoo et al., 2024]. The second is the AI security literature, which studies system prompts as a *target*: how they can be extracted via adversarial querying, and how such leakage can be mitigated [Yang et al., 2026, Zhuang et al., 2025]. The third, most closely related to this paper, is a contemporaneous governance-studies literature exemplified by Neumann et al. [2026], which systematically reviews how researchers and policymakers *discuss* system-level instructions as governance objects, but does not itself perform a direct, coded comparison of the documents' primary text across labs. None of these three literatures asks the question this paper asks: across labs and product categories, what do these documents actually contain, how are they structured, and what does the variation reveal about differing approaches to AI governance?

This gap matters for at least three reasons. First, from a governance perspective, these documents are among the only concrete, inspectable artifacts through which the public can observe how a lab operationalizes its stated values — more concrete than a blog post, more current than a research paper. Second, from an NLP research perspective, treating governance documents as training or evaluation data (as some labs already do internally) requires a shared vocabulary for what these documents contain, which does not yet exist in the public literature. Third, from a practical engineering perspective, developers building on top of these models routinely conflate categorically different artifacts — for instance, comparing a leaked *deployed* prompt for one product against an official *training-time* document for another — producing comparisons that are not actually commensurable.

Contributions

1. We introduce a three-part taxonomy distinguishing training-time constitutions, runtime authority specifications, and deployed inference-time system prompts as categorically distinct artifact types, and show that failing to distinguish them produces invalid comparisons.

2. We construct a credibility-tiered comparative dataset spanning four major labs and a broad corpus of coding-agent products, coded along eight functional axes (identity framing, safety/refusal instructions, tool-use governance, tone/persona control, output format constraints, injection countermeasures, recency handling, and length).

3. We report empirical findings on cross-lab divergence, including an asymmetry in whether labs publish governance artifacts at all, a case where transparency was reactive rather than proactive, a systematic difference in instruction emphasis between consumer-chat and coding-agent product categories, and — using a publicly available, dated changelog — a directly verifiable measurement of publication cadence for one lab's deployed system prompt (18 revisions over 23 months), rather than relying on a publisher's self-description of its artifact as a "living document."

4. We propose a research agenda for treating behavioral-control artifacts as first-class objects of study.

[1] We analyze the 2025-09-12 revision throughout this paper. OpenAI has continued to revise the Model Spec since (including further updates through at least March 2026); as with all artifacts in this study, we report a dated snapshot rather than a currently-synchronized document, consistent with the credibility-tiering and snapshot-transparency approach described in Section 4.

2 Related Work

2.1 Task-Prompt Taxonomies

The dominant strand of prompt-taxonomy research addresses how to structure task instructions to maximize model performance on downstream benchmarks. The TELeR taxonomy [Karmaker Santu and Feng, 2023] categorizes prompts by turn structure, expression style, level of detail, and role, explicitly for the purpose of standardizing benchmark reporting. Broader surveys of prompt-engineering technique [Sahoo et al., 2024] catalog methods such as chain-of-thought, few-shot exemplification, and retrieval augmentation. A related body of work constructs taxonomies of *prompt defects* — failure modes in task-prompt design drawn from software-engineering and NLP venues [Tian et al., 2025]. All of this work treats the prompt as something a user or developer constructs to accomplish a task; none of it addresses the provider-authored instruction layer that this paper studies.

2.2 Prompt Security and Extraction

A substantial and active literature studies system prompts as confidentiality targets. Early work formalized prompt injection and prompt leaking as distinct attack classes [Perez and Ribeiro, 2022]. Subsequent work has measured leakage empirically across large numbers of deployed applications, finding leakage rates exceeding 80% in some large-scale studies [Yang et al., 2026], and has proposed defenses such as proxy substitution [Zhuang et al., 2025] and encoding-based hardening [Sahu et al., 2026]. This literature's object of study is the *event* of extraction and the *robustness* of the prompt against it, not the semantic content of the prompt as a governance artifact. Our work is complementary: where security research asks whether a prompt can be extracted, we ask what it says once available through legitimate or well-established channels, and what patterns exist across many such documents.

2.3 AI Governance Transparency

A smaller literature and body of industry practice addresses transparency in AI governance more broadly, including model cards, system cards, and published behavioral specifications. OpenAI's Model Spec explicitly positions itself as a transparency artifact governing the "chain of command" between the provider, operators (system-prompt authors), and users [OpenAI, 2025]. Anthropic's constitution is explicitly framed as a document written primarily *to* the model itself, optimized for precision over public accessibility, with public release framed as a secondary transparency benefit [Anthropic, 2026a].

The closest prior work to this paper is a contemporaneous study by Neumann et al. [2026], published at FAccT 2026, which treats system-level instructions as a governance object through a PRISMA-based systematic literature review of 287 papers (yielding an eight-category typology of the *goals* researchers attribute to such instructions, e.g., alignment, stability, auditability) combined with a policy-document analysis of two case studies, the US Executive Order 14319 and the EU General-Purpose AI Code of Practice. Their central finding — that governance frameworks often assume system-level instructions provide stable, predictable behavioral control in ways not consistently supported by the technical literature — is a meta-level claim about the literature and policy discourse *surrounding* these artifacts. Their organizing axis is the *stakeholder hierarchy* through which instructions are issued (a "prompt stack" of foundation-model provider, developer/deployer, and end user). Our study differs in scope and method rather than contradicting theirs: we do not survey what researchers or policymakers claim about system-level instructions, but instead directly and comparatively code the content of the primary governance documents themselves, organized around a taxonomy of *artifact type* (training-time, runtime-authority, and deployed) that is orthogonal to the stakeholder-hierarchy axis used in their typology. Notably, based on the sections and reference list available to us at the time of writing, Neumann et al.'s corpus draws on Anthropic's published deployed system prompts but does not appear to treat Anthropic's training-time constitution as a separately governed artifact type, which is consistent with our taxonomy identifying it as a categorically distinct object from the runtime and deployed layers their stakeholder-hierarchy framing is best suited to describe. We view the two studies as complementary: theirs establishes, at the level of the research and policy literature, that the premise of prompt-based governance is more contested than commonly assumed; ours provides a primary-source comparative account of what the documents underlying that premise actually contain.

3 A Taxonomy of Behavioral-Control Artifacts

Informal comparisons of "system prompts" across labs routinely conflate three categorically distinct artifacts. We propose the following taxonomy, motivated directly by the sources examined in this study.

Training-time constitutions. Documents used during model training to shape weights, typically via constitutional-AI-style methods in which the document generates synthetic training signal (example interactions, response rankings, scenario guidance). Anthropic's constitution is the clearest public instance: it is addressed to Claude as its primary audience, is not injected at inference time, and explicitly avoids prescribing rule-like behavior in favor of reasoning the model is expected to internalize [Anthropic, 2026a].

Runtime authority specifications. Documents that govern how a deployed model should arbitrate between instructions from different sources at inference time, without themselves being a single prompt. OpenAI's Model Spec is the clearest instance: it defines a ranked "principal hierarchy" (provider, operator, user) and explains how conflicts between levels should resolve, but is not itself the text injected into any single conversation [OpenAI, 2025].

Deployed inference-time system prompts. The specific text actually prepended to a conversation for a given product at a given time. This is the artifact most commonly leaked, extracted, or — in xAI's case [xAI, 2026] — proactively published in raw form. It is the most ephemeral of the three, changing frequently with product updates, and the most heterogeneous, since a single lab may maintain dozens of distinct deployed prompts across product surfaces.

This taxonomy is not merely descriptive bookkeeping. Section 5 shows that failing to distinguish these types produces comparisons that are not commensurable, and that the choice of *which* artifact type a lab uses to express a given constraint is itself informative about that lab's governance philosophy.

4 Methodology

4.1 Source Tiering

Because the corpus necessarily includes documents of varying provenance, each entry is tagged with a credibility tier, reported transparently rather than treated as uniformly authoritative:

- **Tier 1 (Official/primary):** Content published directly by the originating lab (e.g., Anthropic's constitution, OpenAI's Model Spec, xAI's raw prompt repository).
- **Tier 2 (Reconstructed, high-consensus):** Content appearing consistently across independent extraction attempts, hosted in long-running, widely-cited public repositories (e.g., the coding-agent prompt aggregation studied in Section 5.5).

- **Tier 3 (Single-source leaks):** Lower-confidence material used only to note the possible existence of a pattern, never as a primary data point.

4.2 Coding Scheme

Each entry in the corpus is coded along eight functional axes: identity framing, safety/refusal instructions, tool-use governance, tone/persona control, output format constraints, injection/jailbreak countermeasures, recency/knowledge-cutoff handling, and document length. This scheme was derived inductively from an initial pass over the corpus rather than imposed a priori, consistent with standard qualitative coding practice for emergent taxonomies [Tian et al., 2025].

4.3 Copyright and Reproduction Policy

No system prompt text is reproduced verbatim beyond short quoted fragments where exact wording is analytically load-bearing. All classification in this paper is performed through structural coding and paraphrase, not reproduction, consistent with treating these as evidentiary rather than literary objects.

4.4 Scope and Corpus

The comparative corpus (Table 1) spans four major labs at the level of official or near-official governance artifacts, and a 25-plus-product corpus of coding-agent system prompts aggregated from a single long-running, high-consensus Tier-2 source [Valbuena, 2026], used to characterize product-category-level patterns rather than individual product claims.

Lab	**Artifact**	**Tier**	**Type (per Sec. 3)**
Anthropic	Constitution	1	Training-time
Anthropic	`claude.ai`/mobile system-prompt changelog	1	Deployed (versioned, 18 dated revisions, Jul. 2024–Jun. 2026)
OpenAI	Model Spec	1	Runtime authority spec
xAI	`grok-prompts` repo	1	Deployed (multiple, raw)
Google DeepMind	Prompting guides	3	None published (developer-facing only)
25+ coding-agent products	Aggregated repo	2	Deployed (per-product)

Table 1: Corpus overview by lab/product category, credibility tier, and artifact type. Anthropic is the only lab in the corpus with both a training-time and a versioned, longitudinally documented deployed artifact publicly available.

5 Findings

5.1 Labs Diverge in Artifact *Type*, Not Just Content

The most basic finding is that labs do not agree on what kind of document should perform behavioral governance. Anthropic's primary public artifact is a training-time document not injected at runtime. OpenAI's primary public artifact is a runtime arbitration framework, not itself a prompt. xAI's primary public artifact is the deployed prompts themselves, in raw form, with no equivalent training-time or arbitration-framework document made public. Google DeepMind has published no artifact in any of the three categories that addresses the company's own behavioral constraints on Gemini; its public documentation is prompting guidance addressed to third-party developers, not a transparency document about Gemini's own governance. A naive comparison of "Anthropic's system prompt vs. OpenAI's system prompt vs. Google's system prompt" is therefore not comparing four instances of the same artifact type, but at best two comparable instances and two absences.

5.2 Convergent Hierarchical Framing, Divergent Justification

Despite this divergence in artifact type, Anthropic and OpenAI converge on a strikingly similar structural device: an explicit, ranked hierarchy of authority. OpenAI's principal hierarchy ranks provider, operator, and user instructions, with higher levels overriding lower ones in conflict [OpenAI, 2025]. Anthropic's constitution specifies a priority order of safety, ethics, compliance, and helpfulness. The two labs differ sharply, however, in how they justify the hierarchy: OpenAI frames it in terms of steerability and developer control, drawing an analogy to function definitions and arguments in a programming language, while Anthropic explicitly frames its equivalent structure in terms of reasoning the model should internalize rather than rules to mechanically apply, and explicitly states a preference for reason-based over rule-based specification on the grounds that rule lists do not generalize to novel situations. This is a substantive difference in governance philosophy expressed through a superficially similar structural device, and would be invisible to an analysis that only asked "does the document specify a hierarchy?" without examining the justificatory framing.

5.3 Transparency Can Be Reactive, Not Only Proactive

xAI's decision to publish Grok's raw system prompts is frequently described in press coverage as a transparency initiative comparable to Anthropic's. The provenance differs materially: xAI's public commitment to publish prompts was announced as a direct response to a May 2025 incident in which an unauthorized internal modification caused the Grok bot on the X platform to produce off-topic, ideologically directed outputs [xAI, 2025]. Publication was explicitly framed as a trust-repair mechanism and paired with a commitment to additional internal review controls, rather than introduced as a proactive philosophical stance of the kind that characterizes Anthropic's or OpenAI's framing. This suggests that "publishes its system prompts" is not a single governance stance but a category that includes both prospective transparency-as-values and reactive transparency-as-incident-response, with potentially different implications for how much a given publication should be trusted as representative of typical behavior going forward. Consistent with this, at least one public issue thread on xAI's repository documents a case where a prompt reportedly shared directly by the model in conversation was not subsequently found in the public repository, raising unresolved questions about repository completeness that this paper does not attempt to adjudicate but flags as a limitation of Tier-1 status not implying completeness.

5.4 Substantive, Not Just Structural, Divergence

Labs also diverge in the actual content of behavioral constraints in ways that reflect differing product positioning rather than differing communication style. xAI's published safety-prefix file for its consumer-facing model states that no additional content restrictions apply to fictional adult sexual content involving dark or violent themes unless specified elsewhere, and separately instructs the model to treat users as adults and avoid moralizing about "edgy" requests, with these instructions marked as taking the highest priority over other instructions [xAI, 2026]. This is a substantively more permissive default stance on mature fictional content than is publicly documented for the other labs in this corpus, while the same published material maintains a narrower, explicitly named restriction on content involving minors. The combination — broad default permissiveness paired with one hard, explicitly carved-out exception — appears to be a deliberate product-differentiation choice rather than an oversight, and is stated plainly enough in the primary source that it requires no inference on our part.

5.5 Product Category Predicts Instruction Emphasis

Coding-agent products (Cursor, Claude Code, GitHub Copilot, Devin, and over twenty others aggregated in a single high-consensus Tier-2 source) show a systematic pattern distinct from consumer chat assistants: heavy emphasis on tool-orchestration governance (sequencing of tool calls, confirmation requirements before destructive file operations, diff-format output conventions) and comparatively light emphasis on general content-safety framing. We interpret this as consistent with the three-artifact taxonomy in Section 3: where the underlying base model already carries general safety behavior from a training-time layer (a constitution or equivalent), the deployed, product-specific system prompt's marginal job is to govern what is specific to that product's task domain, not to re-specify general safety behavior the base model already encodes. This is consistent with, though not proof of, a layered division of governance labor across the three artifact types.

5.6 Publication Cadence Reveals Living-Document Status Empirically, Not Just Rhetorically

Both the Anthropic constitution and the OpenAI Model Spec describe themselves in their own text as living documents subject to revision. Anthropic is the only lab in this corpus for which that claim can be verified against a public, dated record rather than taken on the publisher's word: its deployed `claude.ai`/mobile system-prompt changelog lists 18 distinct dated revision events across roughly 23 months (July 2024 to June 2026), a mean interval of approximately 41 days, or under six weeks, between documented changes [Anthropic, 2026b]. The changelog further states explicitly that these updates do not propagate to the API, confirming — independent of any inference on our part — that the deployed prompt is a product-surface-specific artifact rather than a single canonical text shared across all of a lab's surfaces, consistent with the taxonomy in Section 3. No equivalent longitudinal record is publicly available for OpenAI, Google DeepMind, or xAI at the time of writing, which limits comparison of update cadence across labs to this single case; we report it not as a cross-lab comparison but as an existence proof that at least one lab's "living document" framing is independently verifiable rather than solely rhetorical, and as a template for what verifiable transparency could look like elsewhere in the corpus.

6 Discussion

The findings above suggest two things. First, treating "system prompt comparison" as a well-posed question requires first answering *which artifact type* is being compared, and being explicit when a comparison spans types (e.g., a leaked deployed prompt against an official training-time document) rather than presenting it as apples-to-apples. Second, the variation observed across labs is not merely stylistic. The choice of artifact type, the decision whether to publish anything at all, and the circumstances under which publication occurs (proactive versus incident-driven) are themselves meaningful data about a lab's governance posture and should be treated as such by researchers and by the broader public discourse that currently tends to flatten "does lab X publish its prompts" into a single binary marker of trustworthiness.

6.1 Implications for Practitioners

Developers building on top of multiple providers' APIs, including within agentic systems that call more than one model, should be aware that the "safety behavior" of a product is not uniformly encoded at the same layer across providers, and that a system prompt inherited from one product's convention (e.g., a coding-agent's typically light safety framing) should not be assumed portable to a different deployment context without re-examining what layer is expected to carry that responsibility.

6.2 Implications for Researchers

We suggest that behavioral-control artifacts be treated as a legitimate, citable object of NLP and AI-governance research in their own right, with the credibility-tiering approach used here as one candidate methodological norm for handling the mix of official and reconstructed material that necessarily characterizes this domain until more labs adopt full proactive publication.

7 Limitations

This study has several limitations that should inform how its findings are weighted. First, the corpus is not exhaustive: it covers four major labs in depth and one aggregated coding-agent source, not the full space of deployed LLM products. Second, Tier-2 and Tier-3 material, by construction, carries lower evidentiary weight than Tier-1 material, and conclusions drawn substantially from Tier-2 sources (notably the coding-agent product-category finding in Section 5.5) should be read as suggestive pending independent replication against official sources where labs choose to publish them. Third, most artifacts studied here are single or few-point snapshots rather than longitudinal series; the constitution and Model Spec documents are both explicitly described by their publishers as living documents subject to revision, but only Anthropic's deployed-prompt changelog (Section 5.6) provides public, dated, longitudinal evidence of that revision in practice. Findings about relative emphasis or structure for OpenAI, Google DeepMind, and xAI may therefore not hold at a later snapshot, and we have no public basis for estimating how quickly they would change if they did. Fourth, the coding scheme, while derived inductively from the corpus, was applied by a single coder; inter-rater reliability was not assessed and would strengthen future iterations of this work. Fifth, the interpretive claim in Section 5.5 that base-model training absorbs general safety behavior while deployed prompts absorb task-specific governance is consistent with the observed pattern but was not tested against a condition that could falsify it (e.g., a coding agent built on a base model with no equivalent training-time safety layer); we present it as the most parsimonious explanation available from public evidence, not as a demonstrated causal mechanism.

8 Conclusion and Future Work

Behavioral-control documents governing large language models have grown from informal instructions into substantial public artifacts, yet direct comparative content analysis of their primary text — as distinct from literature and policy review of how such documents are discussed [Neumann et al., 2026] — has remained rare. We introduced a three-part taxonomy distinguishing training-time constitutions, runtime authority specifications, and deployed inference-time system prompts, and showed that this distinction resolves otherwise incommensurable comparisons across labs. Applying a credibility-tiered methodology across four major labs and a broad coding-agent product corpus, we found that labs diverge in which artifact type they use at all, that transparency can be reactive rather than proactive, that content divergence reflects genuine product-positioning choices rather than only communication style, that product category systematically predicts which instruction categories a deployed prompt emphasizes, and that at least one lab's claim to maintain a living governance document is independently verifiable against a public, dated record rather than resting on the publisher's word alone.

Future work should extend the corpus to additional labs and product categories as more official material becomes available, assess inter-rater reliability on the coding scheme, and track these artifacts longitudinally given their explicitly living-document status, which would allow the field to study not just cross-sectional divergence but the trajectory of behavioral-control design over time.

References

Anthropic. *Claude's Constitution*. January 2026. https://www.anthropic.com/constitution

Anthropic. *System Prompts* [release-notes documentation, versioned changelog of the deployed claude.ai/mobile system prompt, dated entries from July 2024 through June 2026]. Accessed July 2026. https://docs.claude.com/en/release-notes/system-prompts

A. Neumann, H. Sargeant, and J. Singh. Prompt Governance? On Governing Technologies Governed by Natural Language. *arXiv:2606.07539*, 2026. Also published in *Proceedings of the 2026 ACM Conference on Fairness, Accountability, and Transparency* (FAccT '26), pp. 1–44. https://doi.org/10.1145/3805689.3806763

OpenAI. *The OpenAI Model Spec* (2025-09-12 revision). September 2025. https://model-spec.openai.com/2025-09-12.html

L. Valbuena (x1xhlol). *system-prompts-and-models-of-ai-tools* [GitHub repository, aggregating reconstructed system prompts for 25+ coding-agent products; 141k+ stars, 501 commits at time of access]. Accessed July 2026. https://github.com/x1xhlol/system-prompts-and-models-of-ai-tools

xAI. *grok-prompts* [GitHub repository, official raw system prompts for Grok across chat, X-bot, and API deployment surfaces]. Accessed July 2026. https://github.com/xai-org/grok-prompts

xAI. Statement on unauthorized modification to the Grok response bot on X. Posted to X, May 15, 2025. https://x.com/xai/status/1923183620606619649

S. K. Karmaker Santu and D. Feng. TELeR: A General Taxonomy of LLM Prompts for Benchmarking Complex Tasks. *arXiv:2305.11430*. Accepted to Findings of EMNLP 2023.

P. Sahoo, A. K. Singh, S. Saha, V. Jain, S. Mondal, and A. Chadha. A Systematic Survey of Prompt Engineering in Large Language Models: Techniques and Applications. *arXiv:2402.07927*, 2024.

H. Tian, C. Wang, B. Yang, L. Zhang, and Y. Liu. A Taxonomy of Prompt Defects in LLM Systems. *arXiv:2509.14404*, 2025.

F. Perez and I. Ribeiro. Ignore Previous Prompt: Attack Techniques For Language Models. *arXiv:2211.09527*, 2022. Presented at the NeurIPS 2022 ML Safety Workshop.

Y. Yang, C. Fu, T. Zhang, R. Zeng, Q. Li, T. Du, Z. Wang, S. Ji, and W. Chen. Understanding and Mitigating Prompt Leaking Attacks in Real-World LLM-Based Applications. *arXiv:2606.18673*, 2026.

Z. Zhuang, M.-I. Nicolae, H.-P. Wang, and M. Fritz. ProxyPrompt: Securing System Prompts against Prompt Extraction Attacks. *arXiv:2505.11459*, 2025.

A. Sahu, D. Samanta, and R. Soosahabi. Automated Framework to Evaluate and Harden LLM System Instructions against Encoding Attacks. *arXiv:2604.01039*, 2026.